Sabee Grewal

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Education

In Progress **Ph.D. in Computer Science** *University of Texas at Austin.*
Advisor: Scott Aaronson

2021 **M.S. in Electrical and Computer Engineering** *University of Texas at Austin.*

2019 **B.S. in Physics** *University of Texas at Austin.*

2017 **B.S. in Electrical and Computer Engineering** *University of Texas at Austin.*

Publications

Author ordering in my field is alphabetical by default (all exceptions below are evident).

- **S. Grewal**, V. Iyer, W. Kretschmer, D. Liang. Efficient Learning of Quantum States Prepared With Few Non-Clifford Gates II: Single-Copy Measurements. *In Submission.*
- **S. Grewal**, V. Iyer, W. Kretschmer, D. Liang. Efficient Learning of Quantum States Prepared With Few Non-Clifford Gates. *In Submission.* [arXiv]
- **S. Grewal**, V. Iyer, W. Kretschmer, D. Liang. Improved Stabilizer Estimation via Bell Difference Sampling. *In Submission.* [arXiv]
- S. Aaronson, **S. Grewal**. Efficient Tomography of Non-Interacting-Fermion States. *18th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC 2023), Leibniz International Proceedings in Informatics (LIPIcs)* 266, 12:1–12:18 (2023). [arXiv] [doi]
- **S. Grewal**, V. Iyer, W. Kretschmer, D. Liang. Low-Stabilizer-Complexity Quantum States Are Not Pseudorandom. *14th Innovations in Theoretical Computer Science Conference (ITCS 2023), Leibniz International Proceedings in Informatics (LIPIcs)* 251, 64:1–64:20 (2023). **ITCS 2023 Best Student Paper.** [arXiv] [doi]
- D. E. Koh, **S. Grewal**. Classical Shadows With Noise. *Quantum:* 6, 776 (2022). [arXiv] [doi]

Experience

Research Experience

2021–present **Graduate Research Assistant** *Quantum Information Center*, UT Austin.

summer 2020 **Quantum Computing Research Intern** *Zapata Computing*, Boston, MA.

summer 2019 **Quantum Computing Research Intern** *Zapata Computing*, Boston, MA.

Software Engineering Experience

fall 2020 **Senior Software Engineer** *Zapata Computing*, Boston, MA.

2017–2019 **Software Engineer II** *Microsoft Corporation*, Redmond, WA.

summer 2016 **Software Engineer Intern** *Microsoft Corporation*, Redmond, WA.

summer 2015 **Software Engineer Intern** *Microsoft Corporation*, Redmond, WA.

summer 2014 **CPU Design Intern** *Intel Corporation*, Austin, TX.

Talks & Presentations

2023 **Efficent Tomography of Non-Interacting-Fermion States**.
18th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC 2023)

2023 **Low-Stabilizer-Complexity Quantum States Are Not Pseudorandom**.
14th Innovations in Theoretical Computer Science (ITCS 2023)

Teaching

Fall 2023 **Teaching Assistant** *Quantum Information Science I*, UT Austin.
Fall 2021 **Head Teaching Assistant** *Introduction to Computing Systems*, UT Austin.
Spring 2021 **Teaching Assistant** *Combinatorics and Graph Theory*, UT Austin.
Fall 2019 **Head Teaching Assistant** *Introduction to Computing Systems*, UT Austin.
Spring 2019 **Teaching Assistant** *Software Testing*, UT Austin.
Fall 2016 **Teaching Assistant** *Engineering Programming Languages*, UT Austin.
Fall 2015 **Teaching Assistant** *Introduction to Computing Systems*, UT Austin.
Fall 2014 **Teaching Assistant** *Software Design and Implementation*, UT Austin.

Skills

Programming/Tools: Python, Java, Scala, C++, Bash, LaTeX, Git, AWS, Azure

Awards

2023 **Best Student Paper Award** *14th Innovations in Theoretical Computer Science (ITCS 2023)* .
2020 **Teaching Excellence Award** *Cockrell School of Engineering*, UT Austin.
2015 **Teaching Excellence Award** *Cockrell School of Engineering*, UT Austin.
2014 **Teaching Excellence Award** *Cockrell School of Engineering*, UT Austin.
2014 **Distinguished College Scholar**, UT Austin, (for being in the top 4% of the college).

Last updated: August 2023